



QC7820 high-precision GNSS SoC chip Specifications

Revision History		
Version	Date	Describe
V1.0	2024/05/17	New
V1.1	2024/10/22	Modify the furnace temperature curve
V1.2	2024/12/26	Modify the furnace temperature curve

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1 Product Introduction

1.1 Overview

QC7820 is the fourth generation high-precision GNSS navigation and positioning SoC chip launched by ComNav, integrating RF, power supply, baseband and CPU.

The chip is based on a 22nm low-power design, with a built-in dual-core processor, a main frequency of 800MHz, a maximum of 1.2GHz, and supports high-performance floating-point operations. The chip has a total of 1688 channels, built-in broadband signal reception technology, narrowband anti-interference technology, low-power anti-continuous wave interference technology, and supports multi-system joint positioning/orientation technology, inertial navigation technology, ground-based/satellite-based augmentation technology, SBAS technology, and integrity technology, providing users with highly reliable position and attitude information.

The chip supports multiple positioning modes such as RTK, RTD, PPP, PDP, SPP, etc., and is mainly used in high-precision application fields such as surveying and mapping, positioning and orientation, deformation monitoring, lane-level navigation, commercial drones, and ground robot navigation.

1.2 characteristic

QC7820 has the following features:

- SoC design, integrating GNSS baseband, RF, power supply, processor, storage and interface circuit
- Support single antenna positioning and dual antenna Heading
- Supports all GNSS systems and frequencies
- Supports satellite-based enhanced signals such as SBAS, L-Band, and low-orbit satellites
- Supports 1688 channels, data update rate up to 100Hz
- Based on 22nm process, low power design
- Supports long-distance and highly reliable RTK technology in harsh environments
- Supports broadband RF signal input, built-in narrowband anti-interference module
- Built-in continuous wave elimination module to achieve high-performance anti-continuous wave interference performance at low power consumption

2 Technical specifications

2.1 Technical indicators

Table 1 QC7820 technical parameters

Signal tracking	G PS: L1C/A, L2P, L2C, L5 , L1C
	BDS-2: B1I, B2I, B3I
	BDS-3: B1I, B3I, B1C, B2a, B2b
	G LONASS: G1, G2, G3
	G alileo: E1, E5b, E5a, E5 Alt BoC, E6c
	QZSS: L1C/A, L2C, L5, L1C
	NavIC: L5
	SBAS: L1C/A, L5
	LBand
Positioning and Heading characteristics	Static point positioning: H: 1.5m (1 σ), V: 3m (1 σ)
	Static timing accuracy: 20ns (1 σ)
	Dynamic single point positioning: 20m (1 σ)
	Dynamic speed measurement accuracy: 0.02m/s (1 σ)
	SBAS positioning accuracy: 3D positioning accuracy 1m (1 σ)
	PPP accuracy: Convergence time: 20min, H: 0.1m (1 σ) , V: 0.2m (1 σ)

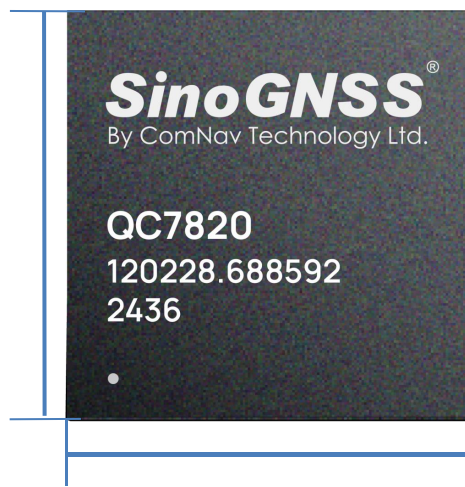
	Pseudorange differential positioning accuracy: H: 0.3m (1 σ), V: 0.5m (1 σ)
	Carrier phase differential positioning accuracy: H: $8\text{mm} + 1 \times 10^{-6} \times D$ (1 σ) V: $16\text{mm} + 1 \times 10^{-6} \times D$ (1 σ)
	Attitude measurement accuracy: Azimuth: $(0.2/R)^\circ$ (R is the baseline length)
RTK Initialization	Initialization time: 5s (baseline length less than 10km)
	Initialization confidence: >99.9%
Time to First Fix	Cold start: 20s
	Hot start: 10s
	Signal recapture: 1s
Data update rate	RTK: 50Hz
	Single point positioning: 50Hz
	Raw data output: 100Hz
Capture/Tracking	High precision mode tracking sensitivity: -155 dbm
	Capture sensitivity: -140dbm
Anti-interference	Anti-narrowband interference signal ratio: 50dB
Data Format	Standard and extended NMEA-0183 (GGA/RMC/GLL/GSV/GSA/ZDA)
	RTCM2.X, RTCM3.X
	BINEX

	PTNL,PJK, PTNL, GGK, PTNL, AVR, NAVPOS, etc.
	Custom binary format ComNav Binary
Communication interface	I2C*1, UART*4, SPI*3, CAN*1, ETH*1
Electrical parameters	Typical power consumption: 0.3W , minimum power consumption: 0.1W
	Working voltage: 3.3V $\pm$ 5%
Environmental characteristics	Working temperature: -40℃—+85℃
	Storage temperature: -55℃—+125℃
	Humidity: 0-100% (non-condensing)
	Static electricity:
	HBM: 2KV
	MM: 200V (After the test, the chip's signal receiving and processing functions are normal)

2.2 Hardware Composition

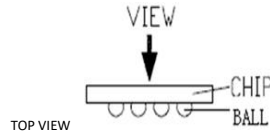
2.2.1 Package size

QC7820 adopts FLIPCHP package and supports two types of packages, large and small. The package size is: 7mmx7mm, and the ball pitch is: 0.5mm.



Size: 7mm×7mm

2.2.2 Pin Definition



	1	2	3	4	5	6	7	8	9	10	11	12	13
A	NC	X_RFIN5	X_RFIN4	RF_VDD_ BG	RF_VDDA _IO1	X_RFIN3	RF_GNDA _RF1	X_RFIN2	X_RFIN1	X_EVENT_I N[0]	X_UART1_ TXD	X_UART1 _RXD	NC
B	RF_GN D_OSC	RF_GND A_RF2	RF_GND A_IF2	RF_GNDA _ADC	RF_VDDA _IO2A	RF_GND A_CP	RF_VDDA _IO2B	X_PPS_ARM	X_PPS2_AR M	X_EVENT_I N[1]	X_ETH_M DC	X_UART2 _TXD	X_UART3 _TXD
C	X_REF_ IN	RF_VSS RF2	RF_SUB_ RF2	RF_SUB_I F2	RF_SUB_ D	RF_GND A_IF1	RF_SUB_I F1	X_VARF_CLK	X_ATOM_CL K	X_ETH_MD IO	VSS_CORE	X_UART2 _RXD	X_UART3 _RXD
D	RF_VD D_IO	RF_VDD _OUT	X_IF_QN	X_IF_QP	X_IF_IP	X_IF_IN	RF_SUB_ RF1	X_CLK_TIC	X_NORMAL _TIC	X_ETH_TX D[1]	X_ETH_TX _EN	X_ETH_TX D[0]	X_UART4 _TXD
E	X_RF_S DI	X_RF_SD O	X_RF_SC LK	X_RF_CF G_MUX	X_RF_CS	X_TCK	X_PWM_ CH0	X_TDI	X_I2C1_SDA	X_ETH_INT D[1]	X_ETH_RX D[1]	X_ETH_R XD[0]	X_UART4 _RXD
F	X_SPI3 _CLK	X_SPI3_ MOSI	X_GPIO[2]	X_GPIO[3]	X_GPIO[5]	X_GPIO[4]	X_TMS	X_TDO	X_I2C1_SCL	X_ETH_RX _ER	X_CAN_R XD	VSS_COR E	VDD_IO
G	X_SPI3 _MISO	X_SPI3_ CSN	X_SPI3_ HOLDN	X_SPI3_ WPN	X_GPIO[1]	X_GPIO[6]	X_GPIO[8]	X_GPIO[7]	X_ETH_CLK	X_ETH_CR S_DV	X_CAN_TX D	VDD_IO	VDD_IO
H	PLL_AV DD	VSS_COR E	X_SPI2_ MOSI	X_SPI2_ MISO	X_GPIO[0]	X_MBIST _TEST	VSS_COR E	VDD_CORE	X_FIQ_TIC	X_ISO_ON _B	X_CAN_ST BY	VSS_COR E	EFUSEIP_ VQPS
J	X_SPI2 _CSN	X_SPI2_ CLK	X_SPI1_C SN	X_SPI2_H OLDN	X_SPI2_ WPN	VSS_COR E	VDD_COR E	VSS_CORE	X_CLK_CON FIG2	X_CLK_CO NFIG1	DCDC_RF_ VSEL1	QLR_GND	QLR_VO UT
K	X_SPI1 _MISO	X_SPI1_ MOSI	X_SPI1_C LK	X_SPI1_ WPN	X_SPI1_ HOLDN	VDD_CO RE	VSS_COR E	DCDC_CORE_ SHUTDOWN	DCDC_CORE _VSENSE	DCDC_COR E_GND	DCDC_CO RE_AVS	DCDC_RF _VSEL0	QLR_AVD
L	VDD_C ORE	VSS_COR E	X_GPIO[10]	X_GPIO[1 3]	X_GPIO[11]	VSS_COR E	VDD_COR E	DCDC_CORE_ VSEL1	DCDC_CORE _AVD	DCDC_COR E_AVS	DCDC_RF_ AVS	ILR_GND	ILR_VOU T
M	X_GPIO [9]	VDD_CO RE	VSS_COR E	X_GPIO[1 5]	VSS_COR E	VDD_CO RE	VSS_COR E	DCDC_CORE_ VSEL0	DCDC_CORE _AVDPWR	DCDC_RF_ VSENSE	DCDC_RF_ GND	X_OSCLIN	ILR_AVD
N	NC	X_HW_R STN	X_GPIO[12]	X_GPIO[1 4]	VDD_CO RE	VSS_COR E	DCDC_CO RE_LX	DCDC_CORE_ LX	DCDC_CORE _AVDPWR	DCDC_RF_ AVDPWR	DCDC_RF_ LX	X_OSCLIO	NC
	1	2	3	4	5	6	7	8	9	10	11	12	13

2.2.3 Pin Marking

Table 2 QC7820 pin description

Location	Signal Name	Function	Type	Valid	Remark
power supply					
L9	DCDC_CORE_AVD		power		3.3V
M9 N9	DCDC_CORE_AVDPWR		power		3.3V
K10	DCDC_CORE_GND		ground		0V
K11 L10	DCDC_CORE_AVS		ground		0V
N7 N8	DCDC_CORE_LX	i500 output voltage	O		Power supply for the chip
K9	DCDC_CORE_VSENSE	Output voltage feedback	I		
K8	DCDC_CORE_SHUTDOWN	i500 shutdown open	I	high	Connect to the enable signal of i500
M8	DCDC_CORE_VSEL[0]	Control the value of i500 output voltage	I		[1][0] 00:0.75v 01:0.80v 10:0.85v 11:0.90v
L8	DCDC_CORE_VSEL[1]	Control the value of i500 output voltage	I		
N10	DCDC_RF_AVDPWR		power		3.3V
M11	DCDC_RF_GND		ground		0V
L11	DCDC_RF_AVS		ground		0V
K12	DCDC_RF_VSEL[0]	Control the value of i100 output voltage	I		[1][0] 00:1.00v 01:1.10v 10:1.20v 11:1.30v
J11	DCDC_RF_VSEL[1]	Control the value of i100 output voltage	I		
N11	DCDC_RF_LX	i100 output voltage	O		Give RF
M10	DCDC_RF_VSENSE	i100 output voltage feedback	I		
M13	iLR_AVD		power		3.3V
L12	iLR_GND		ground		0V
L13	iLR_VOUT	Output 1.8V	O		Power external modules and RF

H13	EFUSEIP_VQPS	Fuse programming high voltage (1.8V)	power		1.8V
A4	RF_VDD_BG		power		1.8V
D1	RF_VDD_IO	1.2~1V	power		1V
A5	RF_VDDA_IO1		power		1V
B5	RF_VDDA_IO2A		power		1V
B7	RF_VDDA_IO2B		power		1V
D2	RF_VDD_OUT		power		RF_VDD_IO
C7	RF_SUB_IF1		power		
C4	RF_SUB_IF2		power		
D7	RF_SUB_RF1		power		
C3	RF_SUB_RF2		power		
C5	RF_SUB_D		power		
B1	RF_GND_OSC	Crystal buffer circuit ground	ground		
A7	RF_GNDA_RF1		ground		
B2	RF_GNDA_RF2		ground		
C6	RF_GNDA_IF1		ground		
B3	RF_GNDA_IF2		ground		
B6	RF_GNDA_CP		ground		
C2	RF_VSS		ground		
B4	RF_GNDA_ADC		ground		
G12 G13 F13	VDD_IO	External capacitor type digital LDO power IO input	power		3.3V
H8 J7 K6 L1 L7 M2 M6 N5	VDD_CORE		power		0.8V(core)
C11 F12 H2 H7 H12	VSS_CORE		ground		0V

J6					
J8					
K7					
L2					
L6					
M3					
M5					
M7					
N6					
JTAG					
F7	X_tms	JTAG TMS, pull up	I		
E8	X_tdi	JTAG TDI, pull up	I		
E6	X_tck	JTAG CLK	I		
F8	X_tdo	JTAG TDO, pull up	O		
SPI					
K3	X_spi1_clk	SPI1 clock signal	IO		Connect to flash
J3	X_spi1_csn	SPI1 chip select signal	IO	Low	
K2	X_spi1_mosi	SPI1 master data output	IO		
K1	X_spi1_miso	SPI1 master data input	IO		
K5	X_spi1_holdn	SPI1 Pause	IO	Low	
K4	X_spi1_wpn	SPI1 write protection	IO	Low	
J2	X_spi2_clk	SPI2 clock signal	IO		Connecting to IMU
J1	X_spi2_csn	SPI2 chip select signal	IO	Low	
H3	X_spi2_mosi	SPI2 master data output	IO		
H4	X_spi2_miso	SPI2 master data input	IO		
J4	X_spi2_holdn	SPI2 Pause	IO	Low	
J5	X_spi2_wpn	SPI2 write protection	IO	Low	
F1	X_spi3_clk	SPI3 clock signal	IO		Reserve
G2	X_spi3_csn	SPI3 chip select signal	IO	Low	
F2	X_spi3_mosi	SPI3 master data output	IO		
G1	X_spi3_miso	SPI3 master data input	IO		
G3	X_spi3_holdn	SPI3 Pause	IO	Low	
G4	X_spi3_wpn	SPI3 write protection	IO	Low	
UART					
A12	X_uart1_rxd	UART1 data input	I		
A11	X_uart1_txd	UART1 data output	O		

C12	X_uart2_rxd	UART2 data input	I		
B12	X_uart2_txd	UART2 data output	O		
C13	X_uart3_rxd	UART3 data input	I		
B13	X_uart3_txd	UART3 data output	O		
E13	X_uart4_rxd	UART4 data input	I		
D13	X_uart4_txd	UART4 data output	O		
I2C					
F9	X_i2c1_scl	I2C1 serial clock line	IO		
E9	X_i2c1_sda	I2C1 serial data line	IO		
GPIO					
H5	X_gpio[0]	Satellite LED(Positing)	IO		I _{OL} =9.5mA I _{OH} =15.4mA
G5	X_gpio[1]	Satellite LED(Heading)	IO		I _{OL} =9.5mA I _{OH} =15.4mA
F3	X_gpio[2]	RTK LED	IO		I _{OL} =9.5mA I _{OH} =15.4mA
F4	X_gpio[3]	reserved	IO		I _{OL} =9.5mA I _{OH} =15.4mA
F6	X_gpio[4]	reserved	IO		I _{OL} =9.5mA I _{OH} =15.4mA
F5	X_gpio[5]	reserved	IO		I _{OL} =9.5mA I _{OH} =15.4mA
G6	X_gpio[6]	reserved	IO		I _{OL} =9.5mA I _{OH} =15.4mA
G8	X_gpio[7]	reserved	IO		I _{OL} =9.5mA I _{OH} =15.4mA
G7	X_gpio[8]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
M1	X_gpio[9]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
L3	X_gpio[10]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
L5	X_gpio[11]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
N3	X_gpio[12]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
L4	X_gpio[13]	FRESET_N (Factory Reset)	IO		I _{OL} =26.2mA I _{OH} =29.2mA
N4	X_gpio[14]	reserved	IO		I _{OL} =26.2mA I _{OH} =29.2mA
M4	X_gpio[15]	reserved	IO		I _{OL} =26.2mA I

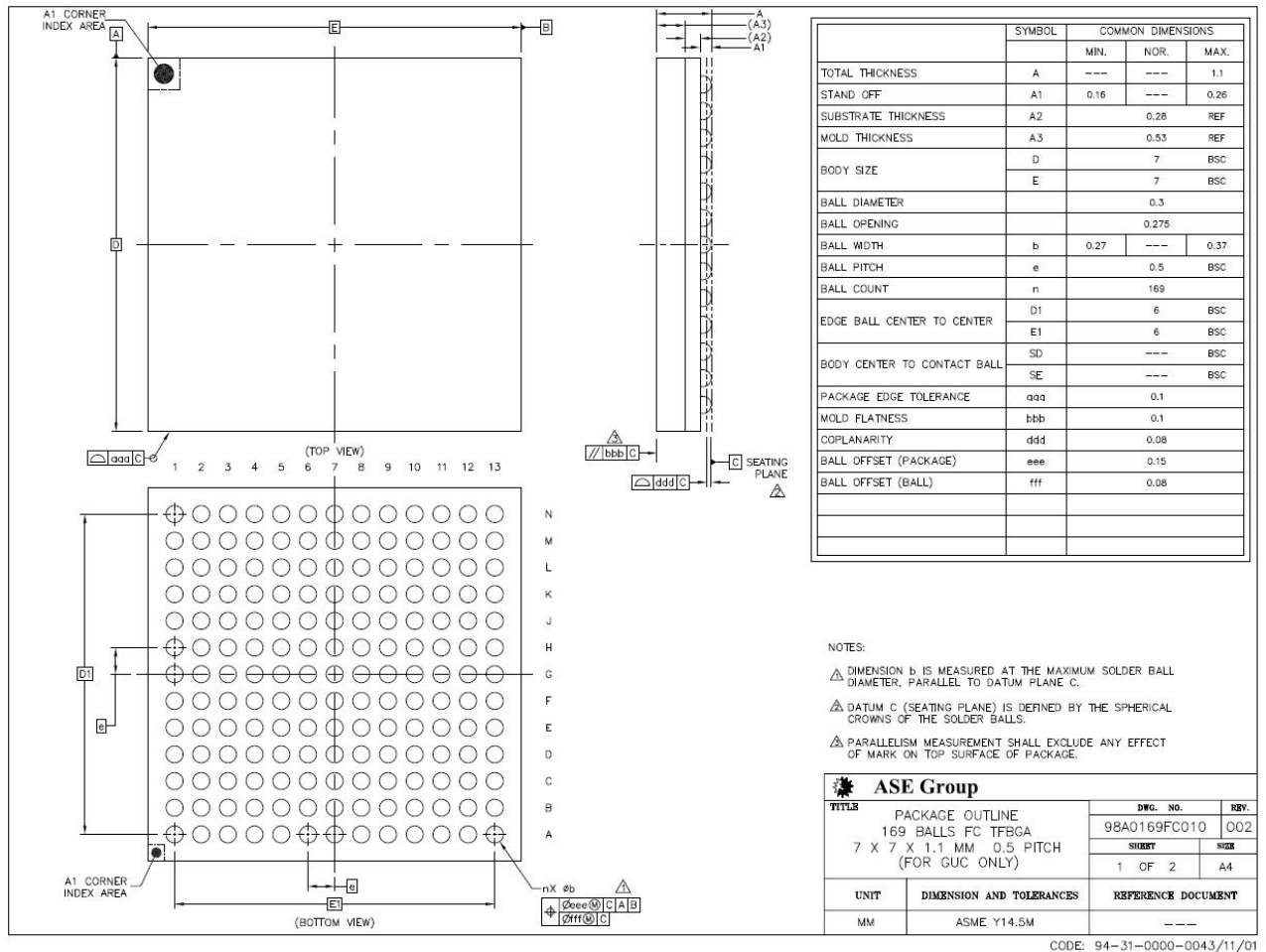
					OH =29.2mA
Baseband extension					
D8	X_clk_TIC	Latch data signal	O		Test foot
D9	X_normal_TIC	Latch data signal	O		
H9	X_fiq_tic	Latch data signal	O		
A10	X_event_in[0]	External event input 0	I		
B10	X_event_in[1]	External event input 1	I		
B8	X_pps_arm	Second pulse output	O		
B9	X_pps2_arm	Second pulse output 2	O		
C8	X_varf_clk	Variable frequency output	O		
C9	X_atom_clk	Atomic clock input	I		
PIT					
E7	X_pwm_ch0	PWM output of PIT channel 0	O		
Reset					
N2	X_hw_rstn	Button reset	I	Low	
CAN					
G11	X_can_txd	Send data to the bus	O		Maximum 5Mbps, CAN transceiver interface voltage 3~5V
F11	X_can_rxd	Receive data from the bus	I		
H11	X_can_stby	Transceiver standby mode enabled	O		
ETH					
E10	X_eth_int	Ethernet interrupt signal synchronized to pclk	O		High-speed interface, IO level reduction helps control EMI
C10	X_eth_mdio	Serial port management interface data input and output	IO		
B11	X_eth_mdc	Serial Management Interface Clock	O		
D10	X_eth_txd[1]	MAC sends data to the transceiver	O		
D12	X_eth_txd[0]	MAC sends data to the transceiver	O		
D11	X_eth_tx_en	Send enable to PHY	O		
E11	X_eth_rxd[1]	The first of two data bits sent by the transceiver on the receive path	I		
E12	X_eth_rxd[0]	Bit 0 of the 2 data bits sent by the transceiver on the receive path	I		
F10	X_eth_rx_er	Receive error signal from	I		

		transceiver			
G10	X_eth_crs_dv	Receives valid data signal from transceiver	I		
G9	X_eth_clk	eth-clock	I		
Clock Configuration					
J10	X_clk_config1 (X_clk_config[0])	Input clock (X_REF_IN) 10M/25M configured to 400M	I		00: Input 25M and configure it to 400M, and you can configure the clock
J9	X_clk_config2 (X_clk_config[1])		I		01: Input 25M is configured as 400M and the clock cannot be configured 10: Input 10M is configured as 400M and can be configured with clock 11: Input 10M is configured as 400M and the clock cannot be configured
RF					
E5	X_RF_CS		I		RF configuration test pin
E3	X_RF_SCLK		I		
E1	X_RF_SDI		I		
E2	X_RF_SDO		O		
E4	X_RF_CFG_MUX	Control signal, when it is low, RF switches to spi signal, when it is high, RF switches to external signal (the above 4 pad signals)	I		Only control CS, SCLK, SDI, SDO signals
A9	X_RFIN1	LNA input (channel 1)	I		RF_VDDA_IO2
A8	X_RFIN2	LNA input (channel 2)	I		
A6	X_RFIN3	LNA input (channel 3)	I		

A3	X_RFIN4	LNA input (channel 4)	I		
A2	X_RFIN5	LNA input (channel 5)	I		
D5	X_IF_IP	IF I analog output positive terminal	O		
D6	X_IF_IN	IF I analog output negative terminal	O		RF_VDD_IO
D4	X_IF_QP	IF Q analog output positive terminal	O		
D3	X_IF_QN	IF Q analog output negative terminal	O		
C1	X_REF_IN	Crystal oscillator reference signal input terminal	I		
RTC					
K13	qLR_AVD		power		3.3V
J12	qLR_GND		ground		0V
J13	qLR_VOUT	External filter capacitor, Powering the crystal	O		0.8V(qLR)
M12	X_osclin	Built-in connection 32.768KHz	I		XA
N12	X_osclio	Built-in connection 32.768KHz	O		XB
H10	X_iso_on_b	RTC Isolation	I		1 means the quarantine is cancelled, 0 means the quarantine is valid
other					
H6	X_MBIST_TEST	Test foot, left to the back end	I		
A1 A13 N1 N13	NC				
H1	PLL_AVD		I		Externally provide 0.8V

2.2.4 Mechanical specifications

Table 3 Mechanical specifications



3 Electrical specifications

The limits given for the QC7820 module are consistent with the maximum ratings (ICE134). Exceeding the limits may cause permanent damage to the device. These are stress ratings only, and operating the device under these conditions or under specifications not given in other sections of this specification may cause damage to the device. Operating under extreme conditions for a long time may affect the reliability of the device.

NOTE: If application information is provided, it is for reference only and does not form part of the specification.

3.1 Maximum Ratings

Table 4 Maximum ratings

Symbol	Parameter	Minimum	Maximum	Unit
VCC	Supply voltage	-0.5	3.6	V
V_BCKP	Backup battery voltage	-0.5	3.6	V
Vin	Input voltage	-0.5	VCC+0.5	V
Ipin	I/O DC Current		TBD	mA
Tsj	Storage temperature	-40	85	°C

Note : The product is not protected against overvoltage or reverse voltage . If necessary , voltage peaks exceeding the supply voltage specifications given in the table above must be limited to within the specified limits by using appropriate diodes .

3.2 Operating conditions

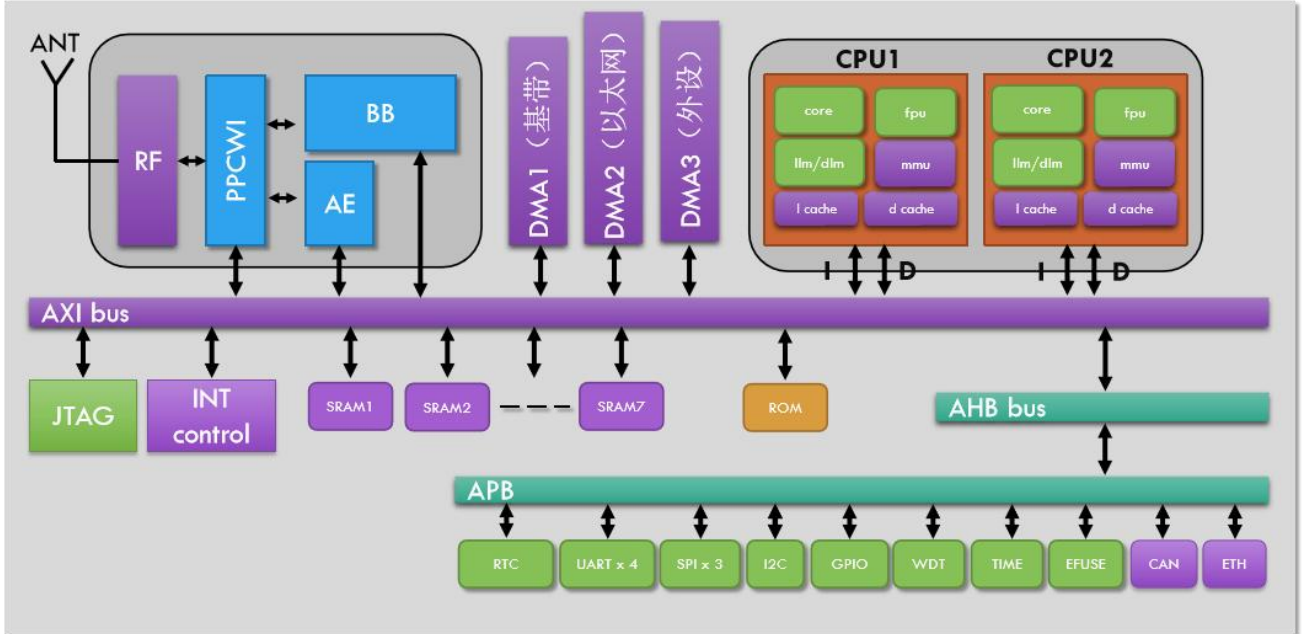
All specifications are based on an ambient temperature of 25 °C. Extreme operating temperatures can significantly affect specifications. Applications operating near these extremes should be tested to ensure that specifications are met.

Table 5 Operating conditions

Symbol	Parameter	Condition	Minimum	Typical	Maximum	Unit
Topr	Operating temperature		-40	+25	+125	°C
VCC	Input voltage range		1.62	3.3	3.63	V
T	Power supply rise time		10		1000	ms
Vreg	Regulating Voltage		0.6		2.5	V
I _{max}	Maximum output current	Corresponding to I100 and I500 respectively			100/500	mA
V _{in}	Digital IO pin low level input voltage		0		0.3	V
V _{il}	Digital IO pin high level input voltage		VCC-0.3		VCC	V
V _{ih}	Digital IO pin low level output voltage		0		0.1	V
V _{ol}	Digital IO pin high level output voltage		VCC-0.1		VCC	V
V _{pad}	Absolute maximum voltage on pad		-0.3		5.5	V
NF _{tot}	Receiver internal noise figure					db

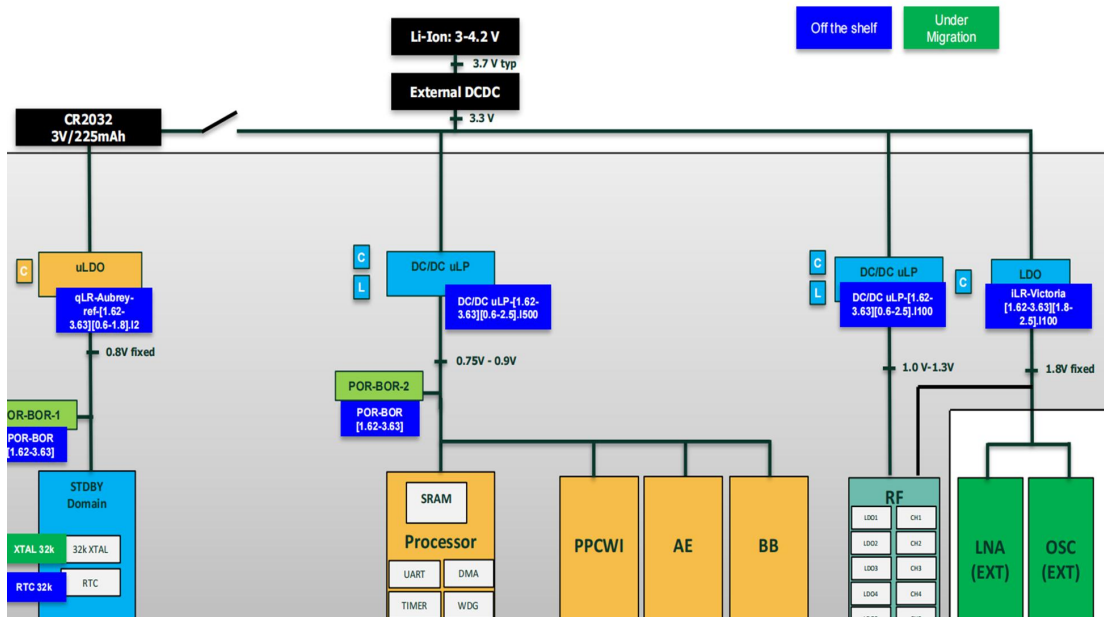
4 Chip Architecture

4.1 Chip frame diagram



4.2 Power Management

4.2.1 System power architecture diagram



4.2.2 External power input

External DCDC supply voltage range 3V-4.2V:

4.2.3 Baseband power domain

The baseband power domain is powered by DCDCi500; DCDCi500 leads two VSEL[1:0] pins to PAD, which can be externally configured to 0.75~0.9V with an accuracy of 0.05V.

4.2.4 RTC power domain

The RTC domain is powered by uLDO (qLR) with an output voltage of 0.8V;

In the always-on domain, uLDO+PORBOR I+XTAL OSC has no control unit.

An externally connected capacitor or rechargeable battery can be charged through the diode after the main power supply is supplied.

4.2.5 RF Power Domain

The RF working condition requires that DCDCi100 and IDO be turned on at the same time to work and provide the main clock for the processor;

DCDCi100 leads out two VSEL[2:1] pins to PAD, which can be externally configured to 1.0-1.3V with an accuracy of 0.1V;

The output of the LDO is 1.8V.

4.2.6 Low voltage reset

PORBOR 1 monitors if the uLDO output voltage is lower than the normal operating voltage of the RTC core and generates a reset signal when it is lower than 0.7V.

PORBOR 2 is the main reset circuit of the chip. It is also used to monitor whether the core power supply voltage is lower than the normal operating voltage and generate a reset signal when it is lower than 0.7V.

4.3 Chip clock

4.3.1 External clock source

- External input clock source X_REF_IN (25M)
- 32.768 kHz low speed external crystal oscillator for RTC function and low power system operation

4.3.2 Internal clock source

The 25M clock gnss_clk generates a 400M processor clock through the processor PLL, allowing the CPU to run at the maximum CPU frequency

4.3.3 Real-time Clock (RTC)

- RTC has an independent clock domain
- RTC clock source includes low-speed external crystal oscillator
- Periodic interrupts: half-second, second, minute, hour, and day interrupts
- Programmable alarm interrupts with specified seconds, minutes and hours values
- Hardware digital trimming to compensate for inaccuracies in external clock sources

4.4 Chip reset

4.4.1 External input reset source

- Button reset (T_rst_b)——T_rst_b is the hard reset interface
- JTAG reset (SRSTn) - JTAG hard reset interface

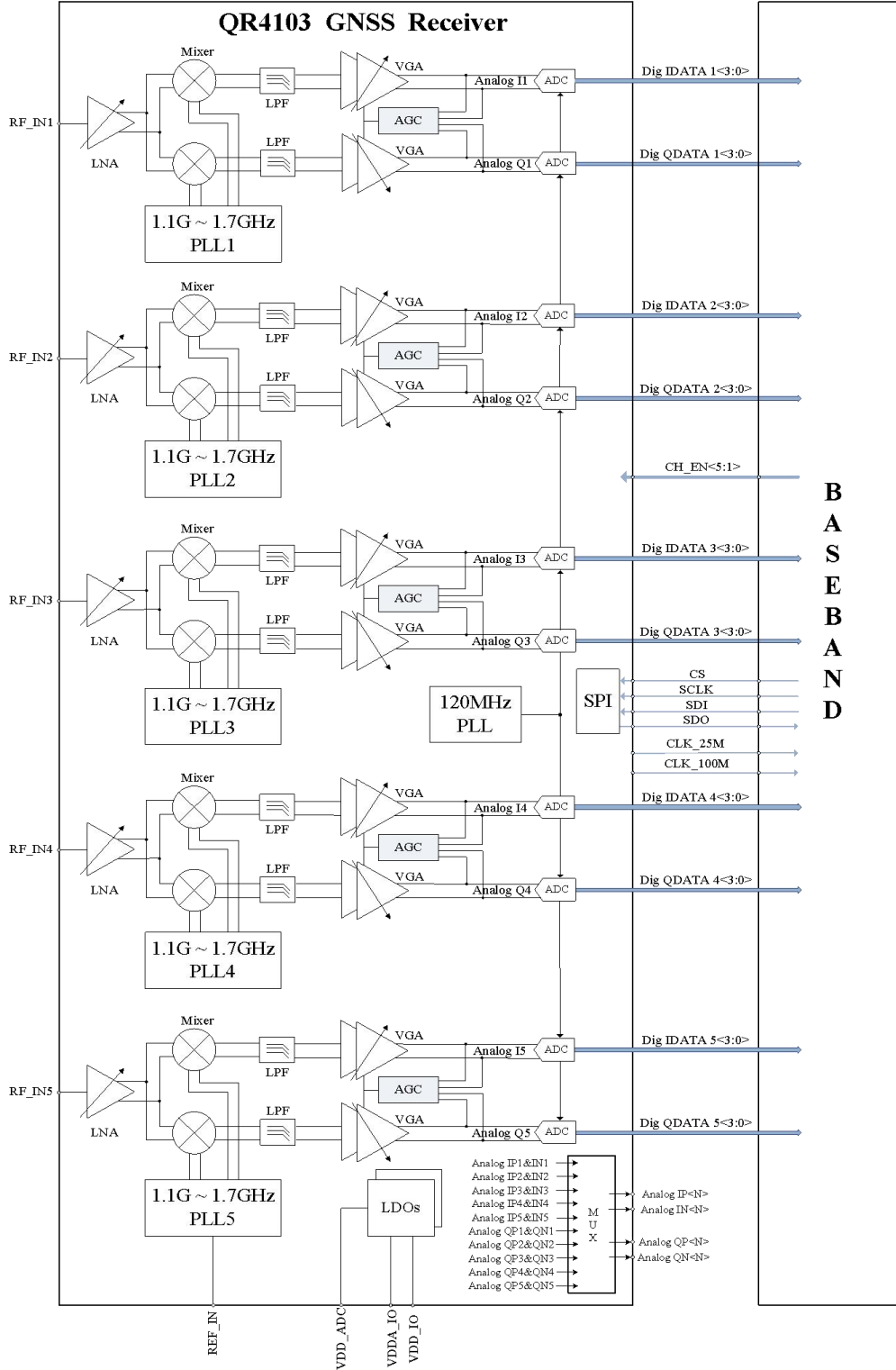
4.4.2 Internal input reset source

- Power-on reset (T_por_n)
- Power-off reset (T_bor)
- wdt (wdt_rst)

- sw (sw_rst)——CPU soft reset interface
- JTAG (edm_srst_req)——JTAG soft reset interface
- Note : Power-on reset (T_por_b) and power-off reset (T_bor) are equivalent ; the reset signal inside the chip is generated after synchronization with the corresponding clock.

5 RF Subsystem

5.1 Functional Block Diagram



5.2 How RF works

The RF subsystem integrates five independent receiving channels, each of which adopts a single-ended input single-conversion image suppression structure design, with a total gain of 26 dB ~ 110 dB , an RF gain adjustment range of 20 dB, an intermediate frequency gain adjustment range of 64 dB , a noise factor of 3 dB , and an image suppression greater than 35 dBc . The receiving channel inputs the GNSS signal, which is amplified by a low-noise amplifier (LNA) and converted into a 0~45 MHz orthogonal intermediate frequency signal using a single-conversion orthogonal mixer. The intermediate frequency signal is then filtered and amplified by a low-pass filter (LPF) and an automatic gain control loop (AGC), and a differential analog intermediate frequency signal with a Vpp of 1V is output. The filter bandwidth can be flexibly configured to 10M ~ 45 MHz .

The frequency synthesizer uses a 2¹ Bits sigma-delta fractional frequency phase-locked loop with a frequency resolution of 10Hz, supports multiple crystal frequencies, and the default reference frequency is 25 MHz . It integrates sub-module circuits such as frequency detector, charge pump circuit, loop filter, divider and voltage-controlled oscillator. PLL provides a stable, high-precision, low-phase noise local oscillator signal for the mixer of the receiving channel. The local oscillators of the five channels are independent of each other and can work normally when configured with the same local oscillator frequency.

The chip integrates a high-performance clock phase-locked loop with a maximum output clock frequency of 120MHz and a clock jitter of 2ps .

5.3 DC parameters

Parameter	Condition	Minimum	Typical Value	Maximum	Unit
Supply voltage	VDD_IO	1	1.5	2	V
Working current	Single Channel		15		mA
	Three channels		45		mA

	Five channels		75		mA
Digital input high level		0.7VDD_IO			V
Digital output high level		VDD_IO-0.4			V
Digital output low level				0.4	V
Operating temperature		-40	27	10 5	° C
ESD	HBM 150pF/1.5KOhm		2		KV

5.4 AC parameters

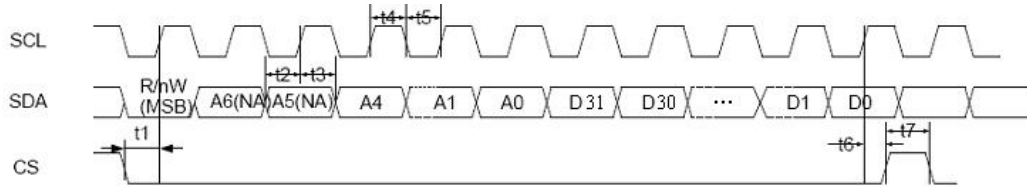
Parameter	Condition	Minimum	Typical Value	Maximum	Unit
RF Input Frequency		1150		1620	MHz
Noise Figure (CASCADED)	DSB (AGC mode)		3	3.5	dB
Input 1dB Compression Point	Gain=40dB (In-band)		-3 5	-3 0	dBm
Spurs at LNA Input	LO leakage		-101		dBm
	Reference harmonics leakage		-103		dBm
Maximum Voltage Gain	Measured from the LNA to the baseband analog output		11 0		dB
IF Analog Output Amplitude	AGC mode (differential outputs)		1	1.2	V p p
I/Q Amplitude Imbalance				±0.5	dB
I/Q Phase Imbalance				±1	degrees
Image rejection		35			dB
VSWR	after matching			1.5	
LNA					
RF Frequency		1150		1620	MHz
Power Gain			30		dB

Gain Flatness	$\pm 10\text{MHz}$		1		dB
Noise Figure			2	2.5	dB
Input 1dB Compression Point			-20		dBm
Reverse Isolation			40		dB
Quadrature Mixer					
RF Frequency Range		1150		1620	MHz
LO Frequency Range		1150		1620	MHz
IF Frequency Range		0		60	MHz
Noise Figure(SSB)			10		dB
In-Band Input 1dB Compression Point			0		dBm
LO Input Power			0		dBm
LO to IN Leakage			-65		dBm
LO to OUT Leakage			-55		dBm
PLL Synthesizer and VCO					
Phase Detector Frequency	Reference frequency division ratio 1~63	1	25	50	MHz
Spurious Signals Due to PFD Frequency			-70		dBc
VCO Output Frequency Range		1150		1620	MHz
Output Power Variation			± 1		dB
LO Phase-Noise Performance	@ $\pm 100\text{Hz}$		-77		dBc/Hz
	@ $\pm 1\text{KHz}$		-88		dBc/Hz
	@ $\pm 10\text{KHz}$		-95		dBc/Hz
	@ $\pm 100\text{KHz}$		-100		dBc/Hz

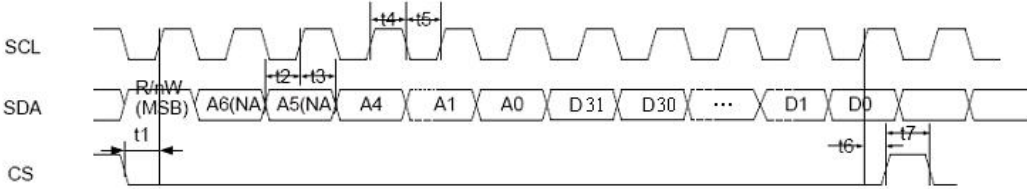
AGC					
Variable Gain Range		0		64	dB
Digital Attenuator Step Size			1		dB
LPF					
Passband 3dB Bandwidth		10/20/25/30/40/45			MHz
In band ripple (Passband Flatness)	0.75 Passband BW		± 1		dB
Stopband Attenuation	measured at 1.5 times of passband BW offset		28		dB
ADC					
Resolution			4		Bits
Integral Nonlinearity			± 0.3		LSB
Differential Nonlinearity			± 0.1		LSB
SNR			25		dB
Differential Input-Voltage Range			1		Vp-p
Input Resistance			5		k Ω
Full-Power Bandwidth	RSOURCE = 50 Ω differential, -3dB rolloff	300			MHz
Conversion Rate		10	25	120	MSPS
Overdrive Recovery Time	$\pm 10\%$ beyond full scale		1		Cycles

5.5 RF Applications

The SPI interface read and write control timing is shown in the following figure:



Write mode



Read mode

Parameter	Description	Min	Type	Max	Unit
t1	CS setup time	20			ns
t2	SDA to SCL setup time	20			ns
t3	SDA to SCL hold time	20			ns
t4	SCL high duration	50			ns
t5	SCL low duration	50			ns
t6	SCL to CS setup time	20			ns
t7	CS pulse width	100			ns

6 Baseband Subsystem

QC7820 chip processor has external interfaces : UART , SPI, I2C , GPIO, WDT, PIT, EFUSE, CAN, ETH, and its configuration is as follows:

6.1 UART

- 4 UART interfaces
- Support hardware handshake to DMA controller
- Supports 8~32 bit oversampling frequency options
- Supports 128 transmit/receive FIFO depth

- Compatible with 16C550A register structure
 - Support 5 to 8 bits per character
 - Supports 1, 1.5 and 2 STOP positions
 - Support parity check and fixed check
 - Support DMA
 - Support programmable baud rate
 - Supports modem controller interface
 - Supports full status reporting function
 - Supports line feed, parity error, framing error and data overflow detection

6.2 SPI

- 3 SPI interfaces
- Complies with AMBA 3 APB protocol specifications
- Support MSB/LSB transmission
- Support DMA
- Support programmable SPI SCLK clock
- Supports memory mapped access via AHB bus or EILM bus (read only)
- Support SPI slave mode
- Configurable Dual and Quad I/O SPI interface
- Configurable TX/RX FIFO depth (128)
- Support 32-bit TX/RX FIFO width
- Configurable programming port location on APB interface
- Mode: slave mode, QSPI mode, SPI direct I/O control

6.3 I2C

- Support AMBA 2.0 APB bus
- Supports standard mode (100kb/s), fast mode (400kb/s) and high speed mode (1Mb/s) protocols
- Programmable master/slave mode
- Supports 10-bit address mode
- Support general call address
- Automatic clock stretching
- Programmable clock /data timing
- Support DMA
- FIFO: 16 bytes

6.4 GPIO

- Supports 16 GPIO channels
- Each channel can be controlled independently
- Programmable I/O direction
- Optional interrupt trigger control support
- Flexible interrupt trigger mode combinations: high/low level trigger and rising/falling/both edge trigger

6.5 WDT

- Support AMBA 2.0 APB bus
- Provides a combination of interrupt and reset when the watchdog timer expires
- Provides write protection mechanism for control/restart registers
- Programmable timer clock source

- Configure magic numbers for write protection registers and restart timers
- The watchdog timer can be paused externally

6.6 PIT

- Four channels 32 bits
- Support AMBA 2.0 APB bus protocol
- Support 4 multi-function timers
- Programmable timer clock source
- Six application scenarios of each multi-function timer (including timer and PWM)
 - 1 32-bit timer
 - 2 16-bit timers
 - Four 8-bit timers
 - 1 16-bit PWM
 - One 16-bit timer and one 8-bit PWM
 - 2 8-bit timers and 1 8-bit PWM

6.7 EFUSE

- Embedded power switch
- Embed four redundant bits
- Power-down and standby modes available
- Asynchronous signal interface
- Macro requirement standard - Vt GO1 (core device) transistor
- TEF22ULP16X32HD18_PHRM for 22nm ULP/ULL 0.8V with 1.8V/2.5V I/O process

- Provides "TEF22ULPESD_P" macro for ESD protection, must be paired with this electrical fuse IP

6.8 CAN

- Embedded power switch
- Embed four redundant bits
- Power-down and standby modes available
- Asynchronous signal interface
- Macro requirement standard - Vt GO1 (core device) transistor
- TEF22ULP16X32HD18_PHRM for 22nm ULP/ULL 0.8V with 1.8V/2.5V I/O process
- Provides "TEF22ULPESD_P" macro for ESD protection, must be paired with this electrical fuse IP

6.9 ETH

- Connected to CPU via AXI bus, 64-bit data bus, 32-bit address bus
- Support 100M/10M
- Supports RMII and PHY connection, but does not support MII mode
- Support DMA sending and receiving
- Support DMA linked list descriptor
- Supports IEEE 1588 for time synchronization (Automotive Ethernet)
- Support MDIO interface management PHY chip
- Support remote wake-up (Wake On Line)
- Support filters (4)

7 Product appearance and logo

7.1 Product Appearance



7.2 Product Identification

First row: SinoGNSS, fixed information, left aligned

Second line: By ComNav Technology Ltd., fixed information, left aligned

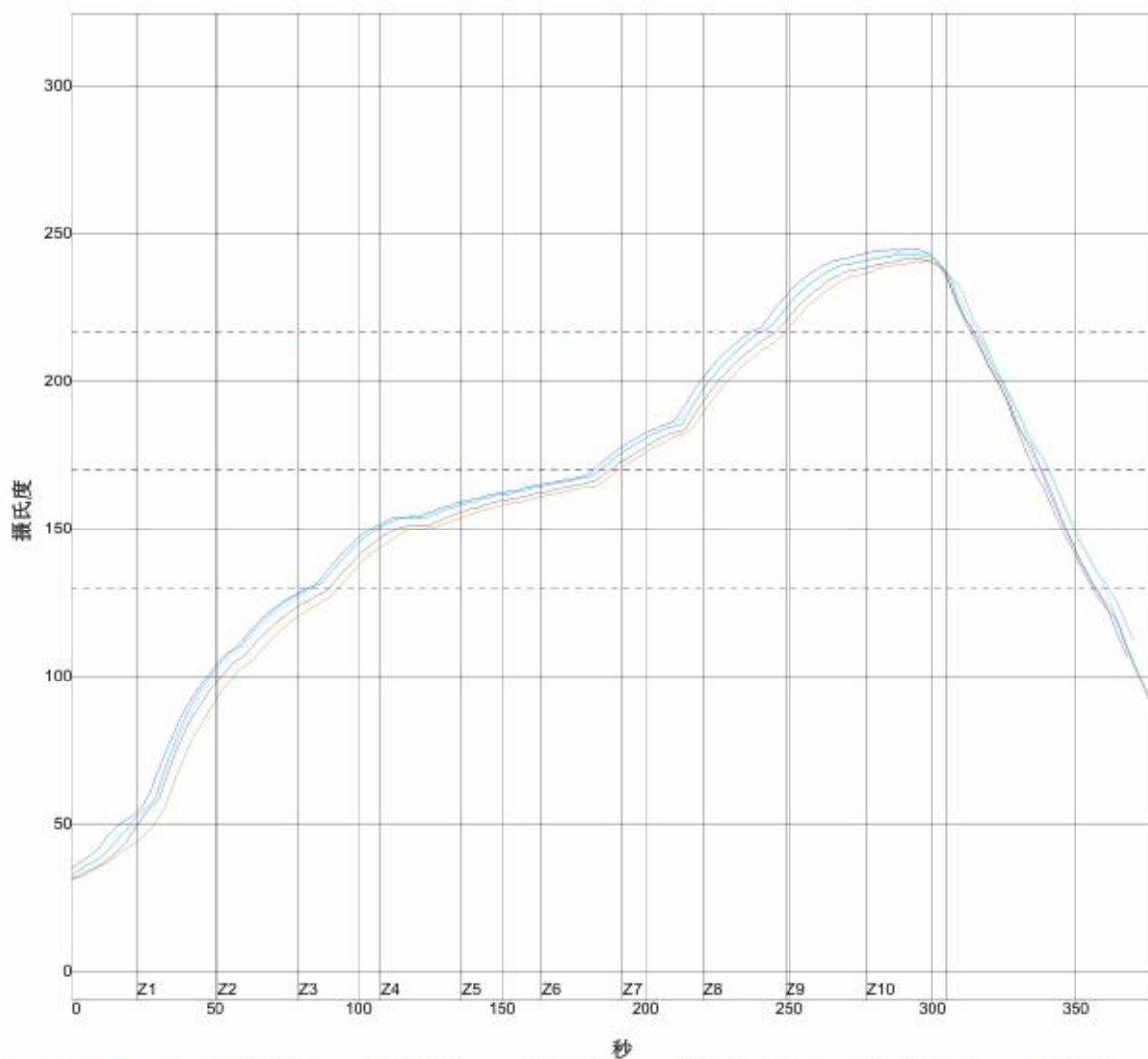
The third line: Product model QC7820, fixed information, left aligned

Fourth line: Product serial number, non-fixed information, left-aligned

The fifth line: product production date, non-fixed information, YY represents year, WW represents week, left-aligned

8 Welding requirements

温度设置 (摄氏度)										
温区	1	2	3	4	5	6	7	8	9	10
上温区	130	140	150	175	175	175	200	245	265	260
下温区	130	140	150	175	175	175	200	245	265	260
传送带速度 (公分/分): 75.0										



PWI= 95%	最高上升斜率	最高下降斜率	预热130至170C	恒温时间170至217C	回温时间 /217C	最高温度
<TC3>	2.49 66%	-2.46 -1%	98.29 28%	55.78 -14%	77.35 -0%	245.16 35%
<TC4>	2.58 72%	-2.36 6%	97.20 24%	58.59 -5%	67.83 -30%	240.83 -22%
<TC6>	2.71 81%	-2.59 -9%	97.87 26%	57.83 -7%	67.94 -29%	241.89 -8%
<TC7>	2.93 95%	-2.68 -15%	100.73 36%	55.88 -14%	74.51 -9%	243.17 9%
<TC8>	2.62 74%	-2.55 -7%	98.62 29%	58.00 -7%	74.67 -9%	243.84 18%
温差	0.44	0.32	3.53	2.81	9.52	4.33

9 Packaging requirements

QC7820 is packaged in trays, with 260 chips in each tray .

Project	Describe
Chip packaging	Anti-static vacuum bag quantity: 1 piece/box
	Humidity card quantity: 1 piece/box
	Desiccant quantity: 1 piece/box
	Number of trays : 1 pair/box (1 tray each)
	Chip quantity: 260 chips/box
	Packing method: vacuum packing
Chip storage	Storage temperature: 23℃±5℃
	Storage humidity: ≤30%